

Cotiviti Practicum – Fall 2026
MS in Biomedical Informatics w/ emphasis in Data Science and AI.
Tara Allred

Timeline: August 24th- December 11th 2026

Practicum Requirements: 180-200 hours

Cotiviti Team Meetings: Thursday's at 130pm MST.

U of U Breaks : No meetings

- Fall Break: October 10-18th
- Thanksgiving: November 26-29th

Pattern Discovery in Clinical Data

Goal: Extract patterns from longitudinal and temporal datasets, including identifying anomalies to combat fraud, waste, and abuse.

Example student contributions:

- Time-series modeling
- Feature engineering for temporal data
- Detecting anomalies related to fraud, waste, and abuse
- Visualizing longitudinal trends

Initial Dataset

- 2024 CMS Medicare Post-Acute Care Utilization – Inpatient Rehabilitation Facility by Geography and Provider
 - CMS Medicare IRF 2022 and 2023 datasets were added for temporal analysis.

✓Phase 1: Beginning – August 24th – September 11th

- Explore CMS IRF dataset and variables
- Identify payment and utilization patterns
- Explore potential outliers
- Use Cotiviti feedback to determine project direction

✓Phase 2: Data Cleaning and Prep – Sep. 14th - October 2nd

- Clean and audit dataset
- Evaluate suppression and missingness

- Create KEEP / REVIEW / EXCLUDE framework
- Engineer per stay payment and utilization features
- Add additional years for temporal analysis

✓ Phase 3: Temporal and Anomaly Analysis – Oct. 2nd-9th

- Compare 2022–2024 provider data
- Analyze year to year payment changes
- Apply anomaly detection and develop provider volume peer groups
- Identify unusual provider payment trajectories
- Evaluate volume, risk and therapy as possible explanations
- Finalize research question and hypothesis

FALL BREAK: OCT-10th-18th – NO CLASS OR MEETINGS

Phase 4: Candidate Investigation – Oct. 19th - Nov. 20th

- Conduct a targeted literature review on Medicare FWA and anomaly detection
 - The goal is to understand how Medicare FWA and healthcare anomaly detection have been studied, identify common approaches and gaps, and provide context for the practicum findings. This is a focused review rather than a comprehensive systematic literature review.
- Investigate selected candidate providers
- Explore explanations for unusual payment patterns
- Determine where claims level data would be needed

THANKSGIVING BREAK Nov 26th-29th NO CLASS OR MEETINGS

Phase 5: Final Project – Paper and Presentation – Nov 30th - Dec 11th

- Refine and validate findings and compare results with literature
- Develop Tableau visualizations
- Evaluate limitations and bias
- Finalize interpretation and conclusions